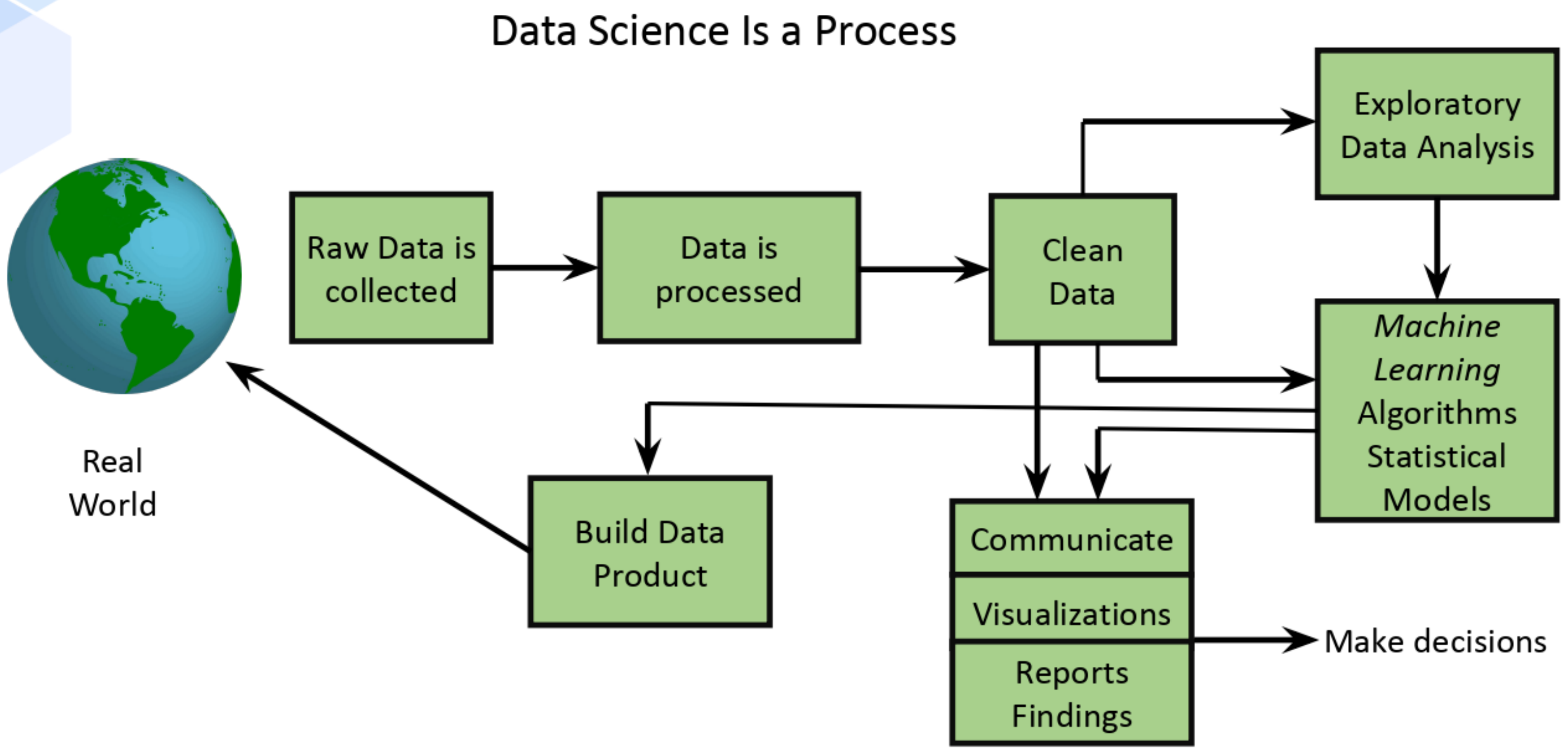
Logistic Regression

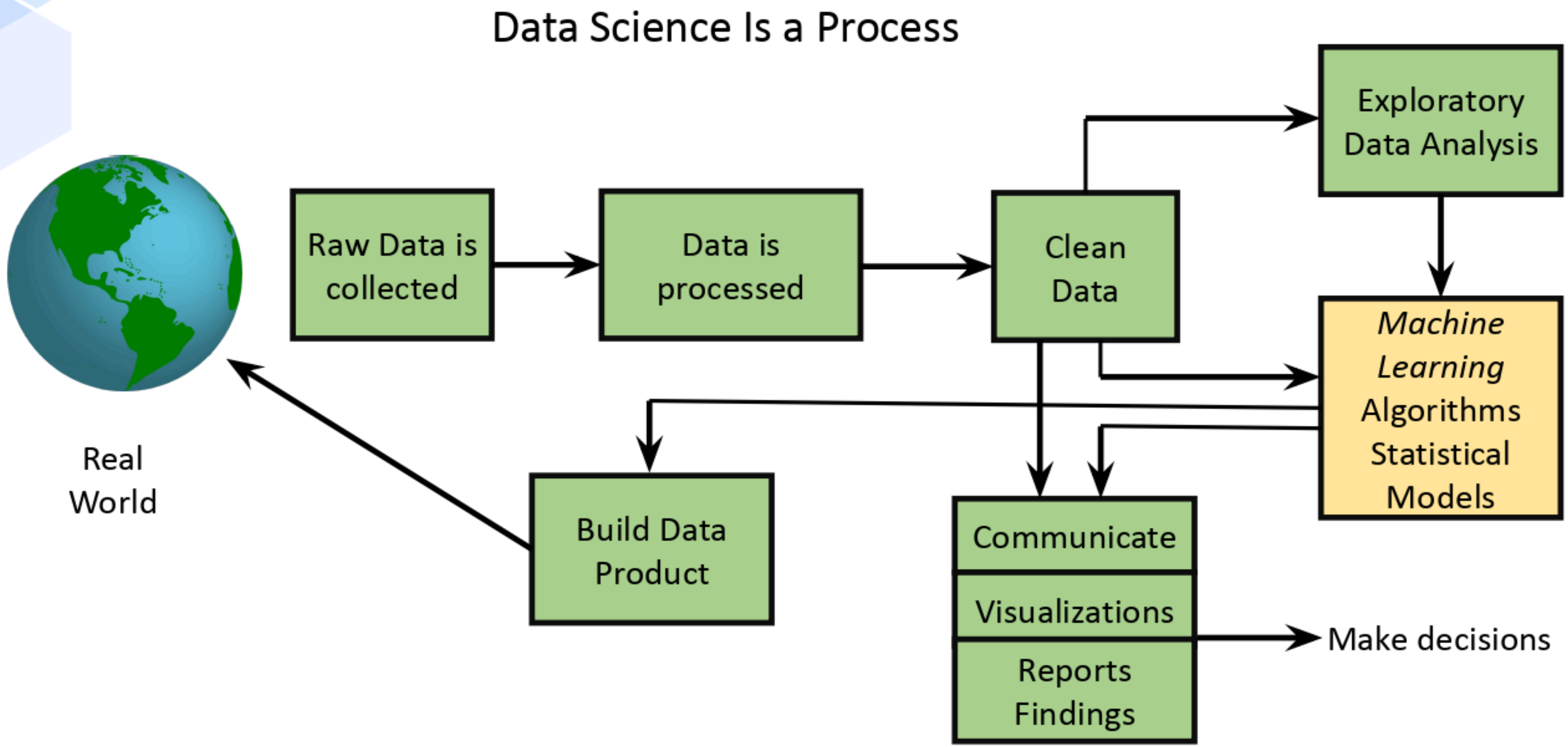
Édipo Cremon

What is Data Science?



O'Neil, Cathy, and Rachel Schutt. *Doing data science: Straight talk from the frontline.* " O'Reilly Media, Inc.", 2013.

What is Data Science?



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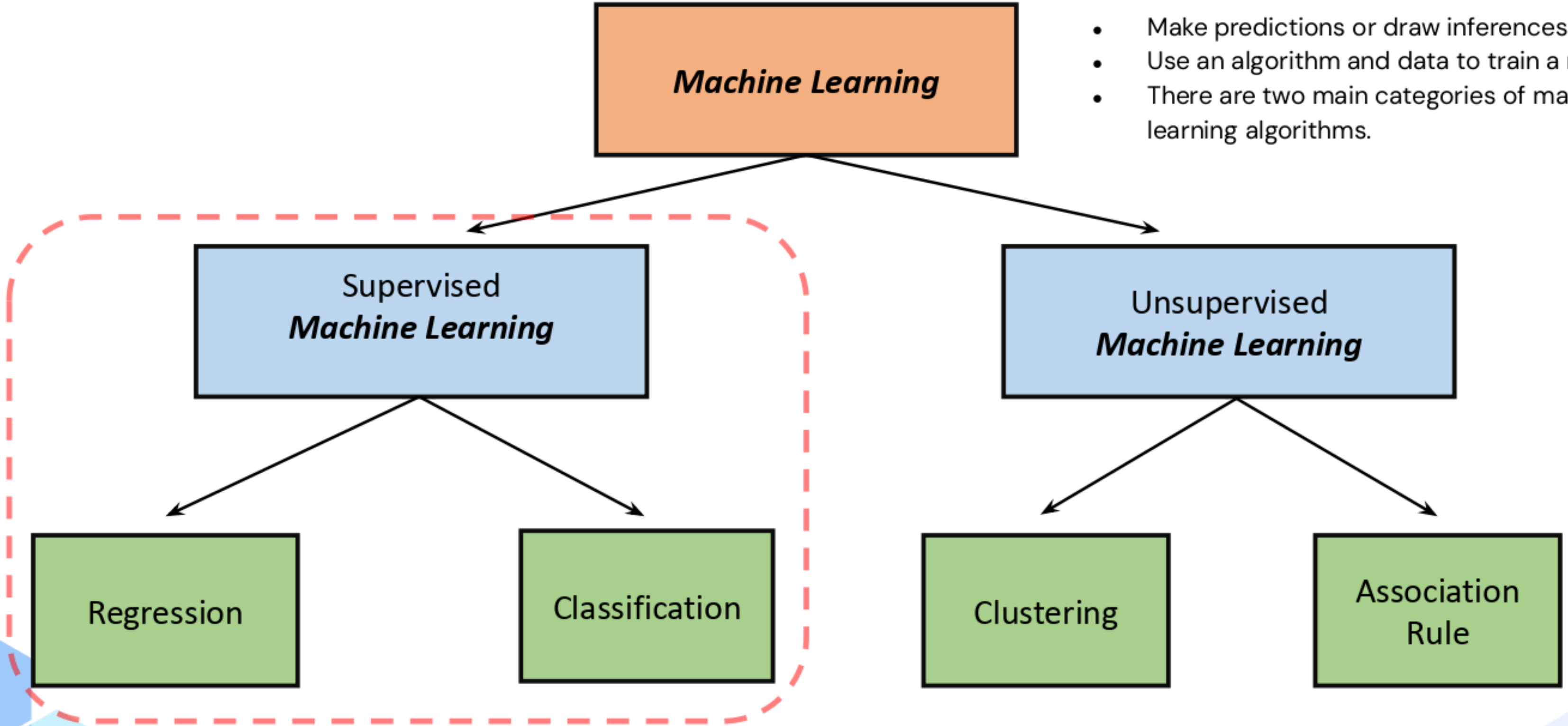
Machine Learning Models



- Make predictions or draw inferences.
- Use an algorithm and data to train a model.
- There are two main categories of machine learning algorithms.

When to use classification instead of regression

Choose the right model based on your target variable

What is your target variable?

Continuous value?

Category or label?

Regression

Predict a continuous numeric value. Exemples: price, temperature, sales, time, etc

Classification

Predict a category or class. Exemples: fraud vs. not fraud, disease vs. healthy, etc

If the goal is to assign a label or group, use classification.
If the goal is to predict a number, use regression.

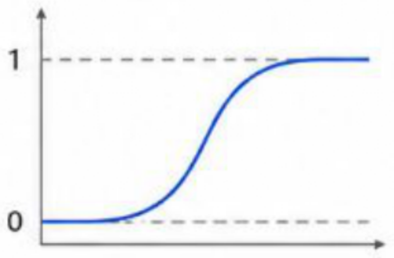
What is logistic regression?

A statistical model usedor binay classification that estimates the **probability of an event** based on input features.

Input feature (x)

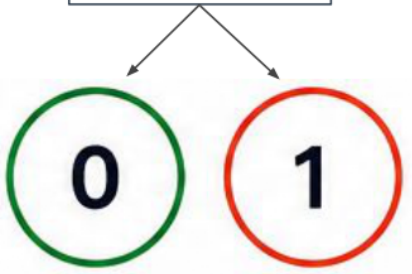
Age
Income
Debt
Payment history

.
.
.



$$\sigma(z) = \frac{1}{1+e^{-z}}$$

P(Y = 1 | X)
Between 0 and 1



Class 0
(No event)

Class 1
(Event)

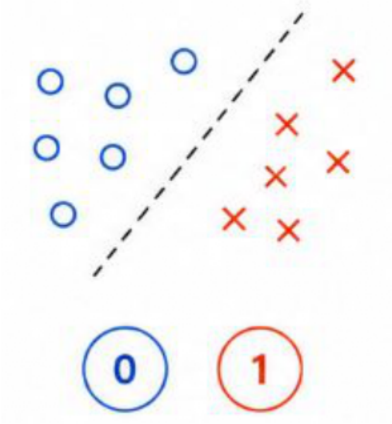
It models the relationship between features and the log-odds of event, then converts it to a probability using the sigmoid function.

Types of classification

In general, the logistic regression can handle different types of categorical target.

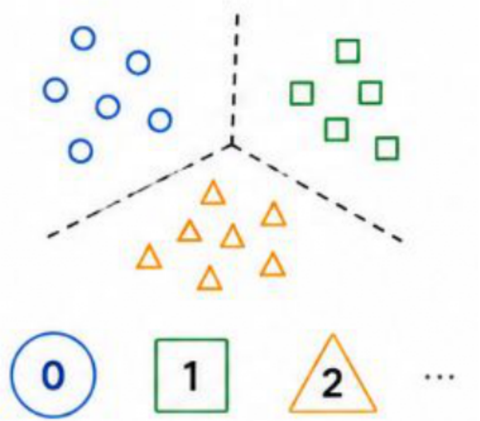
1. Binary classification

Predict one of two classes



2. Multinomial classification

Predict one of three or more unordered classes

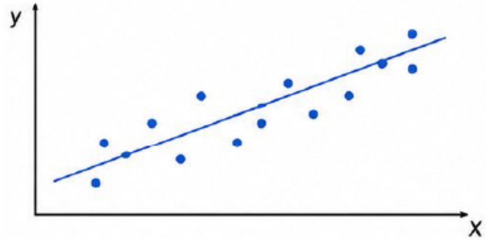


Linear Regression vs. Logistic Regression

Different goals, different outputs.

Linear regression

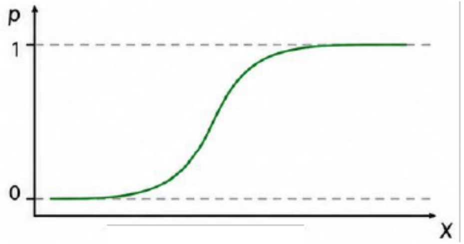
Predict a continuous value.



Output range:
$(-\infty , +\infty)$

Logistic regression

Predict the probability of class (0 or 1).

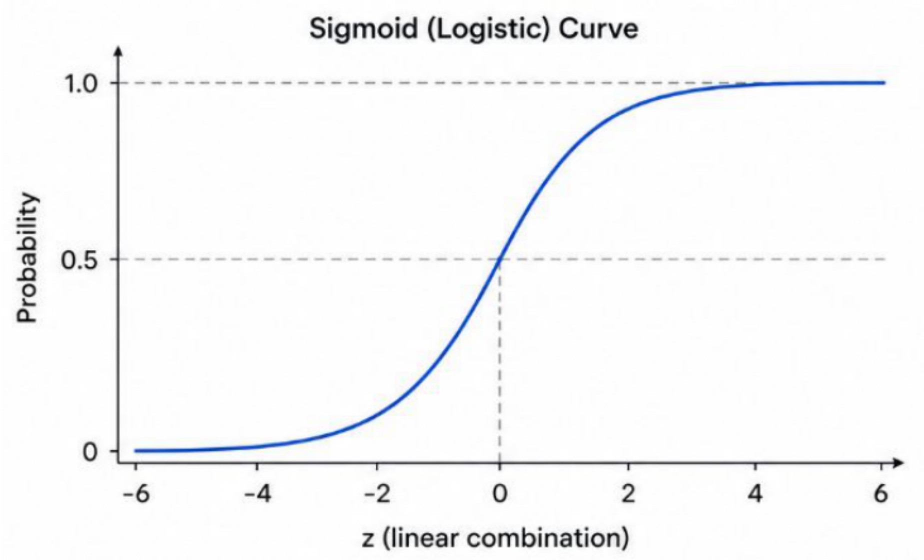


Output range:
$[0, 1]$

Linear relationship	**Model**		S-shaped (sigmoid) relationship
Outputs any real number	**Output**	**0 to 1**	Outputs a probability between 0 and 1
Minimize squared error (OLS)	**Estimation**		Maximize likelihood
Used for regression problems	**Use**		Used for classification problems

The Sigmoid Function

Maps any real-valued number to a probability between 0 and 1.



Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- As z increases, $\sigma(z) \to 1$
- As z decreases, $\sigma(z) \to 0$
- At $z = 0$, $\sigma(z) = 0.5$

Linear combination		Sigmoid function		Probability
$z = w_0 + w_1 x_1 + \ldots + w_p x_p$	→	$\sigma(z)$	→	$P(Y = 1 \mid X) = \sigma(z) \in [0, 1]$

From linear combination to probability

Logistic regression models the log-odds of the event and converts it to a probability using the sigmoid function.

1. Binary classification

Combine features linearly.

$$z = w_0 + w_1 x_1 + \ldots + w_p x_p$$

(can be any real number)

→

1. Binary classification

Model the log-odds of the event.

$$\operatorname{logit}(p) = \ln\left(\frac{p}{1-p}\right) = z$$

$$p = P(Y = 1 \mid X)$$

→

3. Probability

Convert back to probability with the sigmoid function

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$p \in [0, 1]$$

Features (X)

x_1 x_2 $\cdots$ x_p → **Linear Model** (weights $w_0, w_1, \ldots, w_p$) → **Logit** (log-odds) → **Sigmoid** (probability) → $P(Y = 1 \mid X)$ [0, 1]

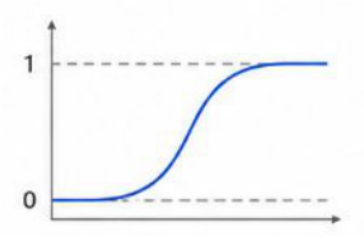
Logistic regression is a linear model on the log-odds scale, which ensures predicted values are valid probabilities between 0 and 1.

Decision boundary and threshold

Logistic regression estimated probabilities. A threshold converts them into class labels.

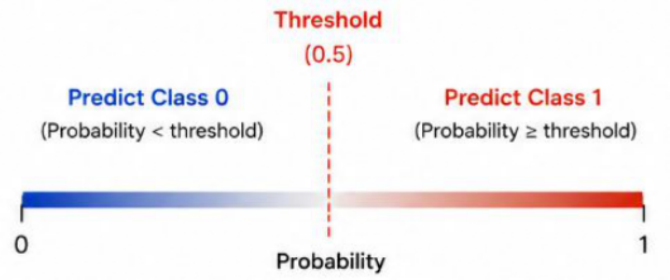
1. Decision boundary

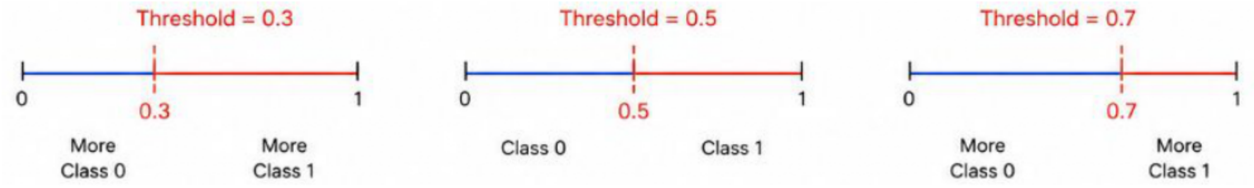
Logistic regression learns a linear boundary that separates the two classes.



2. Threshold on predicted probability

Probabilities are converted to class labels using a threshold (default = 0.5)

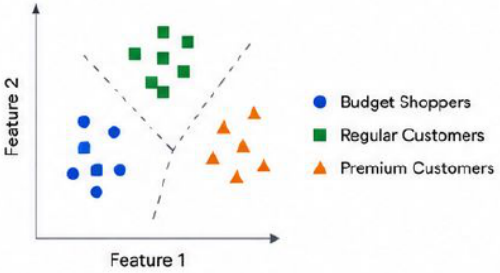




Logistic Regression for multi-class

Example Problem

Predict the type of customer based on age, income, and spending score.



How It Works (Multinomial Logistic Regression)

1. Model the K Classes
For K classes, the model learns K sets of coefficients.

→

2. Compute Scores
For an input x, the model computes a score for each class k.

→

3. Convert to Probabilities
Scores are passed through the softmax function to obtain probabilities for all classes.

→

4. Predict the Class
Predict the class with the highest probability.

Softmax Function

$$P(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

z_k: score for class k

$P(y = k \mid x)$: probability of class k

$$\sum_{k=1}^{K} P(y = k \mid x) = 1$$

Common Strategies

Multinomial Logistic Regression	Directly models all K classes simultaneously using the softmax function. (Preferred when K > 2)
One-vs-Rest (OvR)	Train K binary models (each class vs. all others) and choose the class with the highest probability.

Example Prediction

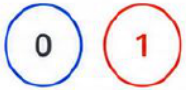
New customer (features) x →

Class	Score (z_k)	Probability $P(y = k \mid x)$
Budget Shoppers	1.50	0.12
Regular Customers	2.30	0.62
Premium Customers	0.70	0.26

→ Predicted Class: **Regular Customers**

Assumptions of Logistic Regression

1. Binary Outcome



The dependent variable is binary (0 or 1).

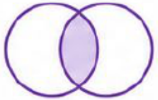
2. Independence of Observations

Observations are independent of each other.

3. Linearity in the Logit

Continuous predictors have a linear relationship with the log-odds.

4. No Multicollinearity

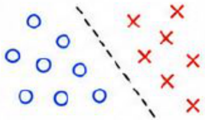


Predictors are not highly correlated with each other.

5. Large Sample Size

A sufficiently large sample improves model reliability.

6. No Perfect Separation



Classes can be separated, but not perfectly.

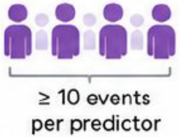
7. Correct Model Specification

All relevant variables are included; irrelevant ones are excluded.

8. Absence of Outliers and Influential Observations

Extreme or influential points should be checked.

9. Sufficient Events per Variable (EPV)



Ensure enough positive events for stable estimates.

Logistic Regression

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